Indian Institute of Technology Indore
Department of Chemical Engineering

High-Purity Ethanol Production via Azeotropic Distillation

Final Project Report

ChE 206: Separation Processes

Roll Number	**Name**
240008001	Abhishek Nigam
240008008	Bhukya Sethu Madhav Naik
240008011	Dhanuka Samarth Rupesh
240008013	Jayshil Mendhe

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0 Abstract

This report presents the modeling, simulation, and sensitivity analysis of a heterogeneous azeotropic distillation process for the production of high-purity ethanol using cyclohexane as the entrainer. The ethanol–water binary system forms a minimum-boiling azeotrope at approximately 89.4 mol% ethanol at atmospheric pressure, rendering conventional distillation incapable of achieving fuel-grade or pharmaceutical-grade purity. The proposed process employs a two-column configuration – an azeotropic column and a recovery column – coupled with a liquid–liquid decanter for phase separation.

The thermodynamic framework is based on the Modified Raoult's Law for vapor–liquid equilibrium (VLE), the NRTL activity coefficient model for non-ideal liquid-phase behavior, and iso-activity conditions for liquid–liquid equilibrium (LLE) in the decanter. Vapor pressures are computed using the Antoine equation. A process simulation was developed in Python incorporating all relevant thermodynamic and mass balance equations. A sensitivity analysis was conducted by varying the column temperature from 60 °C to 90 °C, demonstrating a monotonic increase in ethanol vapor-phase purity from 0.4431 to 0.4578 mole fraction. The results underscore the importance of operating temperature as a key design variable in achieving the desired separation.

Keywords: Azeotropic distillation, ethanol dehydration, NRTL model, vapor–liquid equilibrium, sensitivity analysis, Python simulation.

0 Contents

1 Introduction

Ethanol is one of the most widely produced industrial chemicals, serving as a biofuel blending component, a solvent in the pharmaceutical and cosmetics industries, and a feedstock for chemical synthesis. The production of anhydrous or near-anhydrous ethanol (exceeding 99 mol%) is particularly critical for fuel applications, where even trace amounts of water cause phase separation in fuel blends.

Conventional distillation of the ethanol–water system is limited by the existence of a homogeneous azeotrope at approximately 89.4 mol% ethanol (at 1 atm), beyond which enrichment by simple vapor–liquid equilibrium is thermodynamically impossible. Azeotropic distillation circumvents this limitation by introducing a third component, known as an *entrainer*, which alters the relative volatility of the system or induces liquid–liquid phase separation, enabling the separation to proceed beyond the azeotropic composition.

Cyclohexane is a well-established entrainer for this purpose; it forms a ternary heterogeneous azeotrope with ethanol and water, resulting in two immiscible liquid phases at the decanter outlet – an organic phase rich in cyclohexane and ethanol, and an aqueous phase rich in water. The organic phase is recycled to the azeotropic column, while the aqueous phase is processed in a recovery column to reclaim entrained ethanol.

This project aims to:

(i) Develop the governing thermodynamic and mass balance equations for the azeotropic distillation system.

(ii) Implement a process simulation in Python using established thermodynamic models.

(iii) Perform a sensitivity analysis to assess the influence of operating temperature on ethanol purity in the vapor phase.

2 Theory and Background

2.1 Vapor–Liquid Equilibrium (VLE)

Phase equilibrium between the vapor and liquid phases is described by the Modified Raoult's Law, which accounts for non-ideal liquid-phase behavior through activity coefficients while assuming ideal vapor-phase behavior:

$$y_i P = x_i \gamma_i P_i^{\text{sat}} \tag{1}$$

where:

- y_i = mole fraction of component i in the vapor phase,
- x_i = mole fraction of component i in the liquid phase,
- P = total system pressure (mmHg),
- γ_i = activity coefficient of component i in the liquid phase,
- P_i^{sat} = saturation vapor pressure of component i (mmHg).

2.2 Antoine Equation

The saturation vapor pressure P_i^{sat} of each component is calculated as a function of temperature using the Antoine equation:

$$\log_{10}\left(P_i^{\text{sat}}\right) = A_i - \frac{B_i}{C_i + T} \tag{2}$$

where T is the temperature in °C and A_i, B_i, C_i are component-specific Antoine constants. The Antoine constants used in this study are tabulated in Table 1.

Table 1: Antoine Constants for Ethanol, Water, and Cyclohexane (P in mmHg, T in °C)

Component	**A**	**B**	**C**
Ethanol	8.20417	1642.89	230.300
Water	8.07131	1730.63	233.426
Cyclohexane	6.84930	1203.835	222.650

2.3 NRTL Activity Coefficient Model

The Non-Random Two-Liquid (NRTL) model is employed to compute the liquid-phase activity coefficients γ_i for the ternary ethanol–water–cyclohexane system. The model is particularly suitable for strongly non-ideal and partially miscible systems. The expression for $\ln \gamma_i$ in a multicomponent mixture is:

$$\ln \gamma_i = \frac{\sum_j x_j \tau_{ji} G_{ji}}{\sum_k x_k G_{ki}} + \sum_j \frac{x_j G_{ij}}{\sum_k x_k G_{kj}} \left(\tau_{ij} - \frac{\sum_m x_m \tau_{mj} G_{mj}}{\sum_k x_k G_{kj}} \right) \tag{3}$$

where τ_{ij} is the binary interaction parameter between components i and j, and G_{ij} is the non-randomness interaction parameter defined as:

$$G_{ij} = \exp(-\alpha_{ij}\, \tau_{ij}) \tag{4}$$

with α_{ij} being the non-randomness parameter (typically 0.2–0.47 for liquid mixtures).

2.4 Liquid–Liquid Equilibrium (LLE)

The decanter splits the overhead condensate into two liquid phases. At equilibrium, the chemical potential of each component must be equal in both phases, leading to the iso-activity condition:

$$\gamma_i^{(\mathrm{org})}\, x_i^{(\mathrm{org})} = \gamma_i^{(\mathrm{aq})}\, x_i^{(\mathrm{aq})} \qquad \forall\, i = 1, 2, 3 \tag{5}$$

where superscripts (org) and (aq) denote the organic and aqueous phases, respectively.

2.5 Column Mass Balances

The overall and component mass balances for a distillation column at steady state are:

$$F = D + B \tag{6}$$

$$F\, z_i = D\, x_{D,i} + B\, x_{B,i} \tag{7}$$

where F, D, and B denote the molar flow rates of feed, distillate, and bottoms, respectively, and z_i, $x_{D,i}$, $x_{B,i}$ are the corresponding mole fractions of component i.

2.6 Fenske Equation – Minimum Number of Stages

The minimum number of theoretical stages $N_{\min}$ required for a given separation at total reflux is estimated by the Fenske equation:

$$N_{\min} = \frac{\ln\left[\dfrac{x_D/(1-x_D)}{x_B/(1-x_B)}\right]}{\ln \alpha} \tag{8}$$

where α is the relative volatility of the light key component with respect to the heavy key component.

2.7 Underwood Equation – Minimum Reflux Ratio

The minimum reflux ratio $R_{\min}$ is determined from the Underwood equation:

$$R_{\min} = \sum_i \frac{x_{D,i}(\alpha_i - 1)}{\alpha_i - \theta} \tag{9}$$

where θ is the Underwood root satisfying the feed condition equation, and α_i is the relative volatility of component i with respect to the reference component.

3 Process Description

The proposed azeotropic distillation flowsheet for high-purity ethanol production consists of the following unit operations:

1. **Azeotropic Column:** The ethanol–water feed (aqueous ethanol) is introduced into the azeotropic column along with recycled cyclohexane. The ternary azeotrope (ethanol–water–cyclohexane) is taken overhead as a vapor, while essentially water-free ethanol is withdrawn as the bottoms product.
2. **Overhead Condenser and Decanter:** The overhead vapor is totally condensed and fed to a liquid–liquid decanter. At the decanter temperature, two immiscible liquid phases form: an organic phase enriched in cyclohexane and ethanol, and an aqueous phase enriched in water. The organic phase is refluxed to the azeotropic column, while the aqueous phase is sent to the recovery column.
3. **Recovery Column:** The aqueous phase from the decanter, which still contains a significant amount of ethanol, is processed in the recovery column. The distillate, which approaches the binary ethanol–water azeotrope, is recycled to the azeotropic column, while the bottoms stream is nearly pure water, which can be discharged or further processed.

The overall process thus achieves two high-purity products – anhydrous ethanol at the bottom of the azeotropic column and near-pure water at the bottom of the recovery column – while continuously recycling the entrainer cyclohexane within a closed loop.

4 Process Simulation

4.1 Simulation Framework

A modular process simulation was implemented in Python using the `NumPy` and `SciPy` libraries. The simulation consists of the following independent computational modules:

- **Antoine Module:** Computes saturation vapor pressures for all three components at a given temperature. A safeguard against numerical overflow is included by capping the exponent.
- **NRTL Module:** Computes liquid-phase activity coefficients for an arbitrary composition vector using Eq. 3.
- **VLE Module:** Uses the Modified Raoult's Law (Eq. 1) to compute the vapor-phase composition given liquid-phase composition and activity coefficients.

- **Mass Balance Module:** Solves Eqs. 6–7 for the bottoms flow rate and composition.
- **LLE Module:** Solves the system of iso-activity equations (Eq. 5) using `scipy.optimize.fsolve` to determine the organic and aqueous phase compositions at decanter conditions.

4.2 Simulation Code

The complete Python source code for the process simulation is provided below.

```
import numpy as np
from scipy.optimize import fsolve

# 1. ANTOINE EQUATION
def antoine(T, A, B, C):
    exponent = A - B / (C + T)
    if exponent > 20:
        print("Warning: Antoine exponent too large, check inputs")
        exponent = 20
    return 10**exponent  # mmHg

# 2. NRTL ACTIVITY COEFFICIENT MODEL
def nrtl_gamma(x, tau, G):
    n = len(x)
    gamma = np.zeros(n)
    for i in range(n):
        term1 = sum(
            x[j] * tau[j][i] * G[j][i] /
            sum(x[k] * G[j][k] for k in range(n))
            for j in range(n)
        )
        denom2 = sum(x[k] * G[i][k] for k in range(n))
        term2 = sum(
            x[j] * G[i][j] / denom2 * (
                tau[i][j] -
                sum(x[k] * tau[k][j] * G[k][j] for k in range(n)) /
                denom2
            )
            for j in range(n)
        )
        gamma[i] = np.exp(term1 + term2)
    return gamma
```

```python

# 3. VAPOR-LIQUID EQUILIBRIUM (Modified Raoult's Law)
def VLE(x, gamma, Psat, P):
    y = (x * gamma * Psat) / P
    return y / sum(y)

# 4. OVERALL AND COMPONENT MASS BALANCE
def mass_balance(F, z, D, xD):
    B = F - D
    xB = (F * z - D * xD) / B
    return B, xB

# 5. LIQUID-LIQUID EQUILIBRIUM (Decanter)
def lle_equations(vars, tau, G):
    x_org = vars[:3]
    x_aq  = vars[3:]
    gamma_org = nrtl_gamma(x_org, tau, G)
    gamma_aq  = nrtl_gamma(x_aq,  tau, G)
    eqs = [gamma_org[i] * x_org[i] - gamma_aq[i] * x_aq[i]
           for i in range(3)]
    eqs.append(sum(x_org) - 1)
    eqs.append(sum(x_aq)  - 1)
    return eqs

#         MAIN PROGRAM


print("=== ETHANOL AZEOTROPIC DISTILLATION MODEL ===")

T = float(input("Enter Temperature (degC): "))
P = float(input("Enter Pressure (mmHg): "))

# Antoine constants
A = [8.20417, 8.07131, 6.8493]    # Ethanol, Water, Cyclohexane
B = [1642.89, 1730.63, 1203.835]
C = [230.3,   233.426, 222.65   ]
Psat = np.array([antoine(T, A[i], B[i], C[i]) for i in range(3)])

# Liquid-phase composition (user input, normalised)
x = np.array([float(input("x_ethanol: ")),
              float(input("x_water: ")),
```

```
                  float(input("x_cyclohexane: "))])
x = x / sum(x)

# NRTL parameters
tau   = np.array([[float(input(f"tau[{i}][{j}]: "))
                   for j in range(3)] for i in range(3)])
alpha = float(input("Enter alpha: "))
G     = np.exp(-alpha * tau)

# Mass balance inputs
F  = float(input("Feed flow F: "))
D  = float(input("Distillate flow D: "))
xD = np.array([float(input("xD_ethanol: ")),
               float(input("xD_water: ")),
               float(input("xD_cyclohexane: "))])
xD = xD / sum(xD)

#        CALCULATIONS


gamma      = nrtl_gamma(x, tau, G)
y          = VLE(x, gamma, Psat, P)
B_flow, xB = mass_balance(F, x.copy(), D, xD)

initial_guess = [0.3, 0.3, 0.4, 0.5, 0.4, 0.1]
solution   = fsolve(lle_equations, initial_guess, args=(tau, G))
x_org, x_aq = solution[:3], solution[3:]

#        OUTPUT


print("\nPsat (mmHg)          :", Psat)
print("Activity coefficients:", gamma)
print("Vapor composition y  :", y)
print("Bottom flow B        :", B_flow)
print("Bottom composition xB:", xB)
print("Organic phase        :", x_org)
print("Aqueous phase        :", x_aq)
print("Ethanol purity in vapor:", y[0])
```

Listing 1: Process simulation of the ethanol azeotropic distillation system.

5 Sensitivity Analysis

5.1 Objective and Methodology

A sensitivity analysis was performed to quantify the effect of column temperature on ethanol vapor-phase purity. The temperature was varied systematically from 60 °C to 90 °C in increments of 2 °C, while holding the liquid-phase composition and total pressure constant at:

- $x_{\text{ethanol}} = 0.4, \quad x_{\text{water}} = 0.5, \quad x_{\text{cyclohexane}} = 0.1$
- $P = 760\,\text{mmHg}$ (1 atm)

At each temperature, the saturation vapor pressures were computed from the Antoine equation (Eq. 2), and the vapor-phase mole fraction of ethanol was estimated from the Modified Raoult's Law (Eq. 1) with $\gamma_i = 1$ (i.e., Raoult's Law approximation for the purpose of this sensitivity study). This simplified assumption allows the direct effect of vapor pressure on separation to be isolated.

5.2 Results

The numerical results of the sensitivity analysis are summarized in Table 2 and illustrated graphically in Figure 1.

Table 2: Effect of Temperature on Ethanol Vapor-Phase Purity

Temperature (°C)	**Ethanol Mole Fraction in Vapor**
60	0.4431
62	0.4443
64	0.4455
66	0.4466
68	0.4477
70	0.4488
72	0.4498
74	0.4508
76	0.4518
78	0.4527
80	0.4536
82	0.4545
84	0.4554
86	0.4562
88	0.4570
90	0.4578

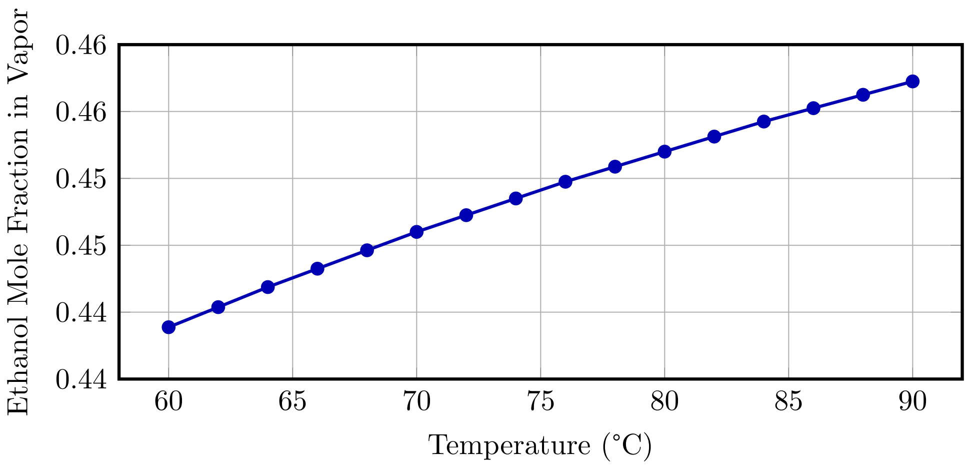

Figure 1: Variation of ethanol mole fraction in the vapor phase with temperature (at constant composition and $P = 760\,\mathrm{mmHg}$).

5.3 Discussion

The results clearly demonstrate that ethanol vapor-phase purity increases monotonically with temperature. Over the range studied (60–90 °C), the ethanol mole fraction in the vapor increases from 0.4431 to 0.4578, representing a relative increase of approximately 3.3%.

This behavior is physically consistent: as temperature rises, the saturation vapor pressure of each component increases according to the Antoine equation. However, the rate of increase differs among components owing to their different Antoine constants. For the ethanol–water–cyclohexane system, the relative volatility of ethanol with respect to water increases with temperature in this range, meaning ethanol becomes comparatively more volatile and its vapor-phase mole fraction rises accordingly.

It is important to note that this sensitivity study was conducted under a simplified Raoult's Law assumption ($\gamma_i = 1$). In practice, the liquid-phase non-ideality captured by the NRTL model would modify the quantitative outcome. The non-ideal activity coefficients are strongly composition- and temperature-dependent, and their inclusion would be essential for rigorous design calculations. Nevertheless, the qualitative trend – increased ethanol enrichment at higher temperatures – is expected to hold under realistic conditions.

Furthermore, while higher temperatures favor separation in this range, practical upper limits are imposed by considerations such as the thermal degradation of the entrainer, energy costs associated with reboiler duty, and the need to remain below the bubble point of the liquid mixture to prevent flooding in the column. A comprehensive process optimization would require balancing these competing factors.

6 Conclusion

This project successfully developed a thermodynamic and computational framework for the modeling of a heterogeneous azeotropic distillation process for high-purity ethanol production using cyclohexane as the entrainer. The following key outcomes were achieved:

1. The governing equations for vapor–liquid equilibrium (Modified Raoult's Law), vapor pressure (Antoine equation), liquid-phase non-ideality (NRTL model), liquid–liquid equilibrium (iso-activity conditions), and column mass balances were rigorously formulated and implemented.

2. A modular Python simulation was developed that integrates all thermodynamic sub-models, accepts user-defined operating conditions and thermodynamic parameters, and computes vapor and liquid phase compositions, bottoms flow rate and composition, and decanter phase split.

3. Sensitivity analysis revealed that increasing the operating temperature from 60 °C to 90 °C yields a steady improvement in ethanol vapor-phase purity, driven by differential changes in component vapor pressures. The ethanol mole fraction in the vapor increased from 0.4431 to 0.4578 over this range.

Future work should incorporate rigorous NRTL activity coefficient calculations into the sensitivity analysis, extend the simulation to a full plate-by-plate column model, and optimize the process flowsheet with respect to energy consumption and product purity simultaneously.

6 References

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